

**Machine Learning models for Diabetes Risk Factors and
Insurance Charges Analyzation**

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Executive summary

This research project delves into the analysis of two datasets: the Diabetes Health Indicator Data and the Medical Cost Personal Dataset. The primary objective is to discern diabetes risk factors for predicting disease occurrence and to understand the determinants influencing individual medical costs. By leveraging these datasets, we aim to contribute to preventive healthcare measures and cost-effective healthcare management.

The Diabetes Health Indicator data, sourced from the Centers for Disease Control and Prevention (CDC), provides a comprehensive overview of health behaviors and chronic conditions among Americans. With 22 variables extracted from over 253,000 entries, this dataset serves as a valuable resource for understanding diabetes risk factors. Similarly, the Medical Cost Personal data, originating from "Machine Learning with R" by Brett Lantz, offers insights into individual demographics and medical insurance charges, facilitating analysis of healthcare expenditure patterns.

In an era where healthcare costs are escalating and chronic diseases like diabetes pose significant health risks, identifying patterns leading to diabetes is crucial. By pinpointing risk factors accurately, individuals can adopt healthier lifestyles and healthcare policies can be tailored to mitigate disease prevalence. Moreover, understanding the determinants of medical insurance charges empowers individuals to make informed financial decisions and allocate resources efficiently for healthcare expenditures.

Through *Principal components analysis (PCA)* and *hierarchical clustering*, we uncovered notable correlations between various health indicators, lifestyle factors, and diabetes occurrence. However, it became evident that no single factor alone is sufficient for precise diabetes prediction. Classification models, including *logistic regression*, *random forest*, and *decision tree*, were employed to assess the predictive capabilities of these factors. While some models exhibited higher precision, they also presented limitations in accurately identifying diabetes cases.

Regression analysis further elucidated significant determinants of medical insurance charges, including age, BMI, smoking status, and region. Leveraging *random forest regression* proved to be effective in predicting charges, offering insights into healthcare cost management. These findings underscore the importance of holistic health management and informed financial planning in navigating healthcare challenges.

In conclusion, this research underscores the critical role of preventive healthcare measures and informed financial decision-making in addressing global health concerns like diabetes. By integrating insights from data analysis, healthcare policies and interventions can be optimized to promote individual well-being and mitigate healthcare costs. Moving forward, it is imperative to continue exploring interdisciplinary approaches to healthcare management and further refine predictive models for enhanced disease prevention and cost-effective healthcare delivery.

1. Introduction

This project centers on the analysis of two datasets: the Diabetes Health Indicator Data (Teboul, 2022) (referred to as the diabetes data) and the Medical Cost Personal Datasets (Choi, 2018) (referred to as the insurance data), both obtained from Kaggle. The primary goal of this research is to identify diabetes risk factors for predicting diabetes occurrence and to comprehend the determinants influencing individual medical costs billed by health insurance, with an aim to reduce these costs.

1.1. Background

Diabetes Health Indicator data

The Centers for Disease Control and Prevention (CDC) conducts the Behavioral Risk Factor Surveillance System (BRFSS), an annual telephone survey focusing on health-related behaviors, chronic conditions, and preventive service usage among over 400,000 Americans (Centers of Disease Control and Prevention, 2014). This survey, initiated in 1984, has been a vital source of data. The Diabetes Health Indicator data utilized in this study comprises 22 variables extracted from 253,680 entries, encompassing details about individuals' fundamental health behaviors and their diabetes status. [Appendix 1]

Medical Cost Personal data

The insurance data originates from the book "Machine Learning with R" by Brett Lantz (Lantz, 2019). This dataset contains 7 variables and 1,338 records, providing general demographics about individuals alongside their corresponding medical insurance charges. [Appendix 2]

1.2. Purpose of Data Utilization:

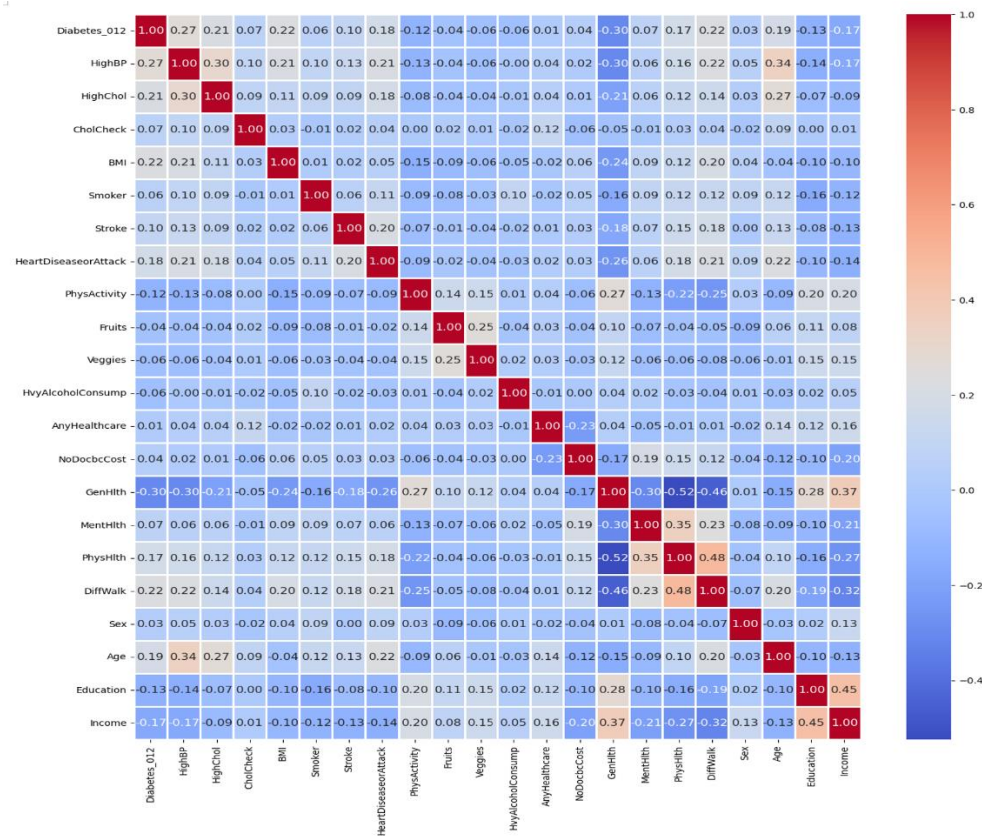
Healthcare remains a prominent topic of discussion, with diabetes being a major global health concern (World Health Organization, 2024). Therefore, by identifying patterns leading to diabetes, more individuals can adopt healthier lifestyles, thereby reducing their likelihood of developing the disease through preventive healthcare measures (Wisner, 2023).

1.3. Research Questions:

1. Which risk factors can most accurately predict diabetes?
2. Which model can accurately predict an individual's risk of developing diabetes?
3. Which model can accurately predict insurance charges?

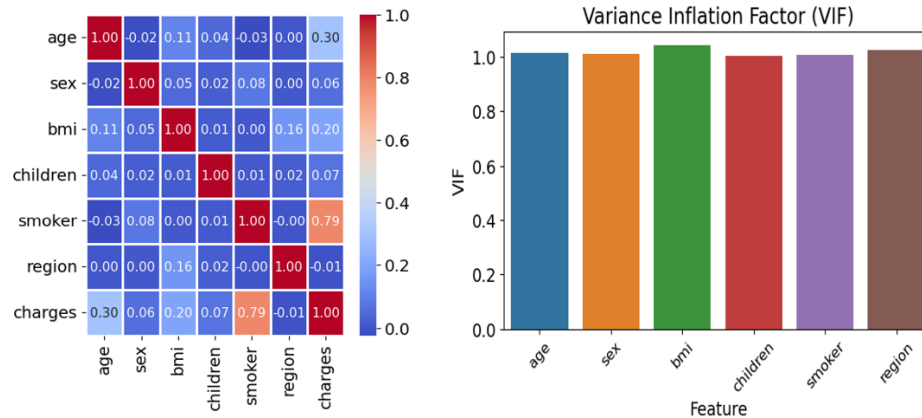
2. Exploratory Data Analysis

Diabetes data



From the correlation pair plot, it's noticeable that general health (0.30), high blood pressure (0.27), BMI (0.22), Difficulty to Walk (0.22), and High cholesterol (0.21) display relatively stronger correlations with the target variable, diabetes, compared to other variables. However, to establish causation between these variables and diabetes, further analysis is required.

Insurance data



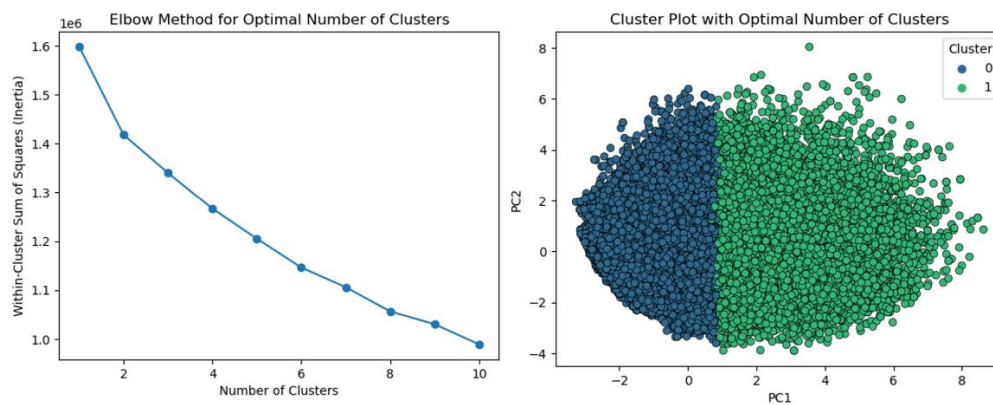
Analysis of the pair plot highlights a notable correlation between smoking status and the target variable, medical insurance charges. Conversely, variables such as region and sex demonstrate relatively weak correlations with medical insurance charges.

Applying Variance Inflation Factor (VIF) analysis following data standardization examine multicollinearity across variables prior to model testing (Frost, n.d.), all variables exhibit VIF index scores below 5, indicating the absence of multicollinearity between variables thereby affirming their utility within the dataset.

3. Research and Findings

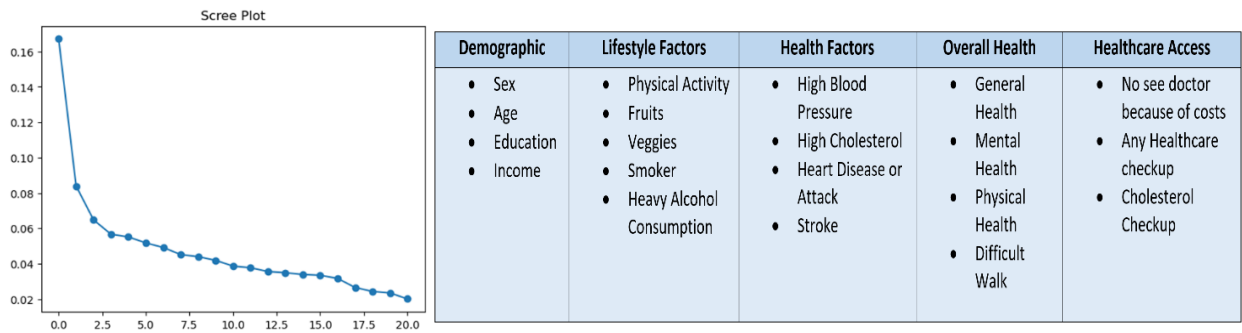
3.1. Unsupervised Learning

3.1.1. Clustering



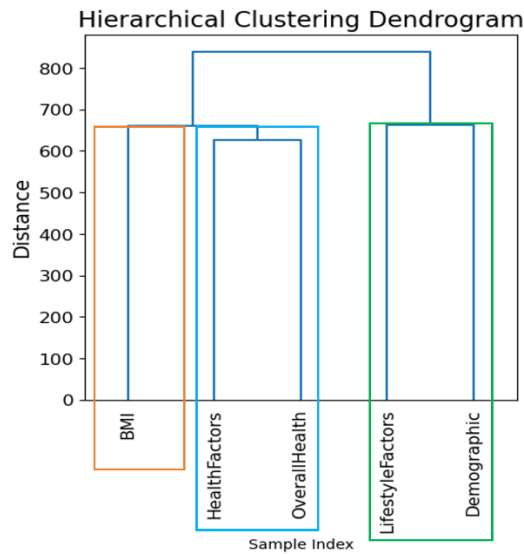
Utilizing the elbow method, we determined that the ideal number of clusters, as indicated by the scree plot, is two. Upon conducting clustering, the scatter diagram illustrates two distinct clusters, differentiating individuals with and without diabetes.

3.1.2. Principal Component Analysis (PCA)



Principal Component Analysis (PCA) was employed to reduce the dimensionality of the dataset to principal components that preserve the most original information (IBM, 2023). Analysis of the scree plot revealed that the first two dimensions adequately capture the data's variance. Consequently, based on PCA results, variables were grouped into subcategories, reflecting potential influencers of diabetes, namely demographic factors, lifestyle attributes, health indicators, overall well-being, and access to healthcare.

3.1.3. Hierarchical Clustering



Subsequently, hierarchical clustering was performed to further delineate subgroups. The analysis revealed three branches at a distance of 650 units. Notably, BMI emerged as a prominent factor, followed by health indicators affecting overall well-being, and lifestyle factors intertwined with demographic characteristics. This hierarchical categorization provides deeper insights into the interrelationships among variables and their collective impact on diabetes susceptibility.

3.2 Classification

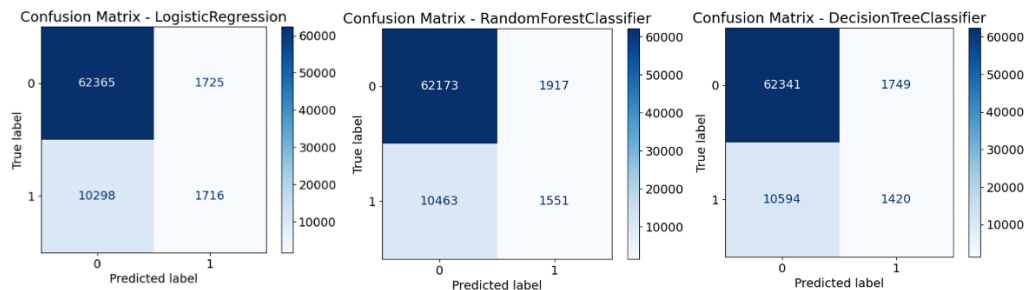
The subgroupings derived from hierarchical clustering, namely health factors, lifestyle factors, and BMI, were subjected to classification models such as logistic regression, random forest, and decision tree to ascertain the primary contributors to diabetes. The evaluation metrics utilized include accuracy, precision, F1 scores, and confusion matrices, which collectively illustrate the models' predictive capabilities in accurately identifying diabetes cases.

3.2.1. Classification result analysis

i) Health factors

Health_with_diabetes:		
Model: LogisticRegression	Model: RandomForestClassifier	Model: DecisionTreeClassifier
Accuracy: 0.84	Accuracy: 0.84	Accuracy: 0.84
Precision: 0.50	Precision: 0.45	Precision: 0.45
F1 Score: 0.22	F1 Score: 0.20	F1 Score: 0.19

For the health factors, all three models achieved an accuracy of 0.84. Notably, logistic regression demonstrated higher precision (0.50) compared to random forest and decision tree (0.45 and 0.44 respectively), with a slightly superior F1 score (0.22). However, despite these metrics, the F1 score suggests that health factors alone are insufficient for precise diabetes prediction across all three models.

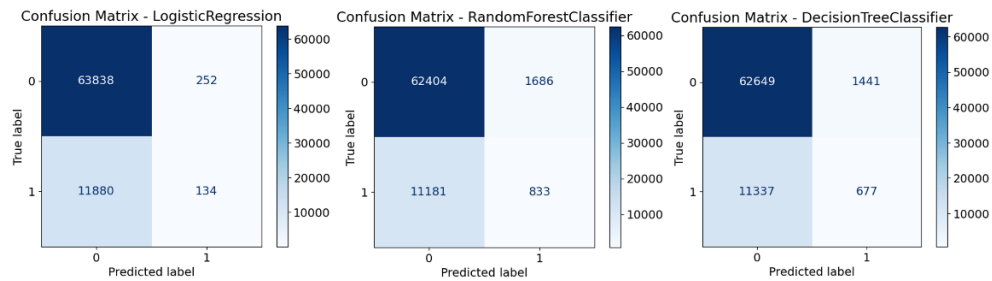


The confusion matrix highlights logistic regression's ability to correctly predict true positive results (62365 cases) and true negative results (1716 cases). Conversely, decision tree exhibited the highest number of false negative results, while random forest presented the most false positive results among the three models.

ii) Lifestyle factors

Lifestyle_with_diabetes:		
Model: LogisticRegression	Model: RandomForestClassifier	Model: DecisionTreeClassifier
Accuracy: 0.84	Accuracy: 0.83	Accuracy: 0.83
Precision: 0.35	Precision: 0.33	Precision: 0.32
F1 Score: 0.02	F1 Score: 0.11	F1 Score: 0.10

In terms of lifestyle factors, logistic regression achieved an accuracy of 0.84, slightly outperforming random forest and decision tree, both with accuracies of 0.83. Logistic regression also showed higher precision (0.35) compared to random forest and decision tree (0.33 and 0.32 respectively), but with a lower F1 score (0.02). Despite these findings, the F1 score indicates that lifestyle factors alone inadequately predict diabetes across all models.

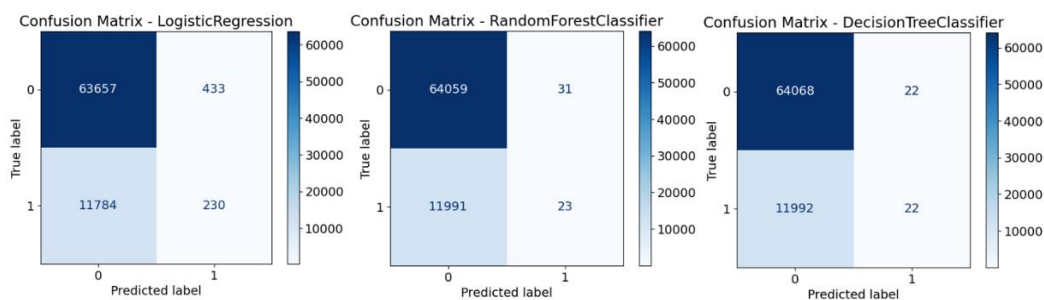


The confusion matrix reveals logistic regression's ability to accurately predict true positive results (63838 cases), while random forest correctly predicts the most true negative results (833 cases). However, logistic regression also presents the highest number of false negative results, alongside random forest.

iii) BMI

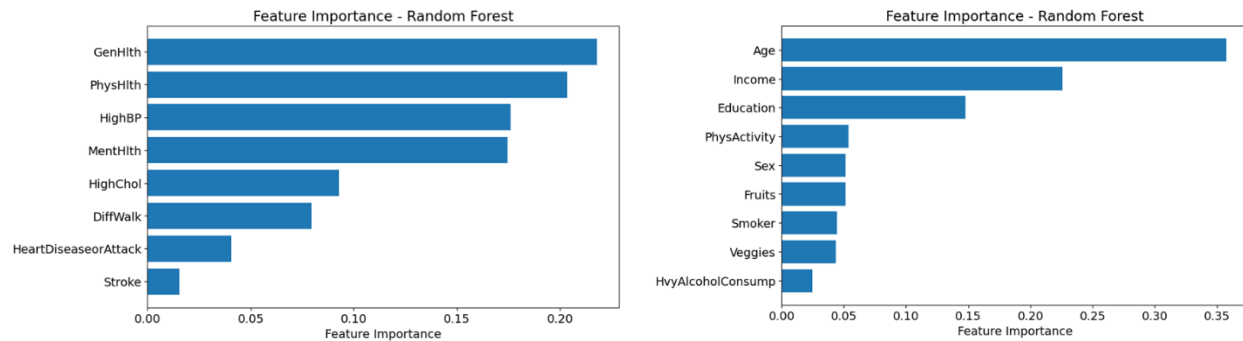
BMI_with_diabetes:	Model: LogisticRegression	Model: RandomForestClassifier	Model: DecisionTreeClassifier
Accuracy: 0.84	Accuracy: 0.84	Accuracy: 0.84	
Precision: 0.35	Precision: 0.43	Precision: 0.50	
F1 Score: 0.04	F1 Score: 0.00	F1 Score: 0.00	

Similarly, for BMI, all three models attained an accuracy of 0.84. Interestingly, decision tree exhibited the highest precision (0.50), surpassing logistic regression and random forest (0.35 and 0.43 respectively). However, logistic regression achieved a marginally better F1 score (0.04) compared to the other models (0.00). Nonetheless, the F1 score suggests that BMI alone is insufficient for precise diabetes prediction across all models.



The confusion matrix illustrates decision tree's ability to accurately predict true positive results (64068 cases), while logistic regression correctly predicts the most true negative results (230 cases). However, decision tree also presents the highest number of false negative results, alongside logistic regression. Notably, in this context, models that effectively predict true positive results also tend to predict false negative results, while those predicting the most true negative results also tend to predict the most false positive results.

3.2.2. Variables importance



Additionally, utilizing random forest to identify variable importance for both health and lifestyle factors revealed predictors for diabetes. Among health factors, general health, physical health, high blood pressure, and mental health emerged as potential predictors, while demographic factors such as age, income, and education were identified as predictors within lifestyle factors.

The classification results reveal that no single factor, including health factors, lifestyle factors, or BMI, can adequately predict diabetes. Although logistic regression showed higher precision, especially with health and lifestyle factors, none of the models achieved consistently high F1 scores. This highlights limitations in accurately identifying diabetes cases using individual factors. Additionally, while decision tree exhibited superior precision in some cases, it also produced more false negatives, suggesting a trade-off between complexity and interpretability.

These findings highlight the potential of combining health and lifestyle factors, along with BMI, for more accurate diabetes prediction. Furthermore, studies have indicated additional socioeconomic factors, such as income and education level, play significant roles in diabetes prevalence. For instance, a 1% increase in per capita income is associated with a 0.02% decrease in diabetes prevalence (Selena E. Richards, 2022), while higher levels of education are inversely associated with diabetes prevalence (Luisa N. Borrell, 2006). Incorporating these socioeconomic factors alongside health and lifestyle indicators could enhance the predictive power of diabetes risk assessment models.

3.3 Regression

OLS Regression Results						
=====						
Dep. Variable:	charges	R-squared:	0.751			
Model:	OLS	Adj. R-squared:	0.750			
Method:	Least Squares	F-statistic:	668.1			
Date:	Mon, 18 Mar 2024	Prob (F-statistic):	0.00			
Time:	16:54:06	Log-Likelihood:	-13548.			
No. Observations:	1338	AIC:	2.711e+04			
Df Residuals:	1331	BIC:	2.715e+04			
Df Model:	6					
Covariance Type:	nonrobust					
=====						
	coef	std err	t	P> t	[0.025	0.975]

const	-1.182e+04	955.130	-12.371	0.000	-1.37e+04	-9941.729
age	257.2881	11.886	21.647	0.000	233.971	280.605
sex	-131.1106	332.811	-0.394	0.694	-784.001	521.780
bmi	332.5701	27.722	11.997	0.000	278.186	386.954
children	479.3694	137.644	3.483	0.001	209.346	749.393
smoker	2.382e+04	411.843	57.839	0.000	2.3e+04	2.46e+04
region	-353.6400	151.927	-2.328	0.020	-651.682	-55.598
=====						
Omnibus:	299.003	Durbin-Watson:	2.088			
Prob(Omnibus):	0.000	Jarque-Bera (JB):	713.975			
Skew:	1.207	Prob(JB):	9.17e-156			
Kurtosis:	5.642	Cond. No.	296.			
=====						

The Ordinary Least Squares (OLS) regression (Alto, 2023) results offer valuable insights into the relationship between various variables and medical insurance charges in the dataset. With an R-squared value of 0.751, approximately 75.1% of the variability in charges can be explained by the variables in the model.

Among these variables, age, BMI, number of children, smoking status, and region exhibit statistically significant coefficients ($p < 0.05$), indicating their significant impact on medical insurance charges. Specifically, each additional year of age corresponds to an increase of approximately \$257.29 in charges, while a unit increase in BMI leads to an increase of about \$332.57 in charges. Smokers tend to have substantially higher medical charges, with an approximate difference of \$23,820 compared to non-smokers. Additionally, individuals with more children incur higher charges, with each additional child contributing to an increase of approximately \$479.37 in charges.

However, the coefficient associated with sex does not appear to be statistically significant ($p > 0.05$), suggesting that this variable may not significantly impact medical charges after considering other factors. Therefore, it will be excluded before undergoing further model evaluation.

3.3.1 Regression Analysis

The regression analysis provides key metrics including Mean Squared Error (MSE), Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and R-squared. MSE, RMSE, and MAE quantify the disparity between predicted and actual results, reflecting the model's accuracy (Hodson, 2022). Meanwhile, R-squared indicates the proportion of variability in the data explained by the model, serving as a measure of its predictive performance (Miles, 2014).

Linear Regression Results:
 MSE: 31998334.01174948
 RMSE: 5656.706993627076
 MAE: 4019.8466711909136
 R-squared: 0.7781260396455787

Lasso Regression Results:
 MSE: 31996413.992298733
 RMSE: 5656.537279316625
 MAE: 4019.6031609914307
 R-squared: 0.77813935291118

Ridge Regression Results:
 MSE: 32002181.04475477
 RMSE: 5657.047025149674
 MAE: 4021.1696437398127
 R-squared: 0.7780993646178044

Random Forest Regression Results:
 MSE: 21327448.224170424
 RMSE: 4618.165027818995
 MAE: 2759.5046301110365
 R-squared: 0.8521171320977832

Among the regression models evaluated, including linear regression, lasso regression, and ridge regression, consistent results were observed with RMSE at 5656.7, MAE at 4019.85, and R-squared value at 0.778. However, random forest exhibited superior performance, demonstrating lower RMSE at 4618.17, MAE at 2759.50, and a higher R-squared value of 0.852. These findings suggest that random forest has the potential to better predict insurance charges from the given variables, as it effectively captures the variance within the data.

Overall, the findings suggest that age, BMI, smoking status, number of children, and region are significant determinants of medical insurance charges. Leveraging random forest regression can enhance the accuracy of predicting charges, thereby facilitating more informed decision-making in healthcare cost management.

4. Conclusion

In light of the analysis of both datasets, understanding the risk factors for diabetes is critical in encouraging preventive health measures. These insights not only enable individuals to adopt healthier lifestyles but also play an important role in determining healthcare policy and insurance programs. Preventive healthcare measures, which are frequently included in medical insurance policies (Craford, 2024), are critical for the early diagnosis and treatment of chronic diseases such as diabetes. By accurately predicting diabetes risk factors, individuals can proactively engage in preventive screenings and interventions, ultimately reducing the burden of healthcare costs and improving overall well-being.

Furthermore, predictive modeling provides insights into medical insurance charges, allowing individuals to make informed financial decisions and prepare for healthcare spending. With escalating healthcare expenses, recognizing prospective charges enables individuals to budget efficiently and obtain adequate insurance coverage.

In essence, prioritizing preventive healthcare measures, alongside understanding medical insurance charges, underscores the importance of holistic health management. While diabetes represents a significant health concern, the broader context of preventive healthcare encompasses various aspects of well-being, contributing to healthier and more financially secure communities.

5. References

- Alto, V. (2023, 2 14). *Understanding Ordinary Least Squares (OLS) Regression*. Retrieved 3 25, 2024, from builtin: <https://builtin.com/data-science/ols-regression>
- Centers of Disease Control and Prevention. (2014, 5 16). *About BRFSS*. Retrieved 3 25, 2024, from Centers of Disease Control and Prevention: <https://www.cdc.gov/brfss/about/index.htm>
- Choi, M. (2018). *Medical Cost Personal Datasets*. Retrieved 3 25, 2024, from Kaggle: <https://www.kaggle.com/datasets/mirichoi0218/insurance>
- Craford, S. (2024, 2 15). *What's Preventive Healthcare Insurance and What's Covered Under These Plans?* Retrieved 3 25, 2024, from healthline: <https://www.healthline.com/health/consumer-healthcare-guide/preventative-health-insurance>
- Frost, J. (n.d.). *Variance Inflation Factors (VIFs)*. Retrieved 3 25, 2024, from Statistics By Jim: <https://statisticsbyjim.com/regression/variance-inflation-factors/>
- Hodson, T. O. (2022, 7 19). Root-mean-square error (RMSE) or mean absolute error (MAE): when to use them or not. *Geoscientific Model Development*, 15(14), 5481 - 5487. doi:10.5194/gmd-2022-64
- IBM. (2023, 12 8). *What is principal component analysis (PCA)?* Retrieved 3 25, 2024, from IBM: <https://www.ibm.com/topics/principal-component-analysis>
- Lantz, B. (2019). *Machine Learning with R, Third edition*. (J. Lovell, Ed.) Livery Place, Birmingham, UK: Packt Publishing Ltd. Retrieved 3 25, 2024
- Luisa N. Borrell, F. J. (2006, 9). Education and Diabetes in a Racially and Ethnically Diverse Population. *Am J Public Health*, 1637 - 1642. doi:10.2105/AJPH.2005.072884
- Miles, J. (2014, 9 29). R Squared, Adjusted R Squared. *Wiley StatsRef: Statistics Reference Online*. doi:10.1002/9781118445112.stat06627
- Selena E. Richards, C. W. (2022, 7 28). Lifestyle and socioeconomic determinants of diabetes: Evidence from country-level data. *PLoS One*. doi: 10.1371/journal.pone.0270476
- Teboul, A. (2022). *Diabetes Health Indicators Dataset*. Retrieved 3 25, 2024, from Kaggle: <https://www.kaggle.com/datasets/alexteboul/diabetes-health-indicators-dataset>
- Wisner, W. (2023, 2 7). *What Is Preventive Health and Why Is It Important?* Retrieved 3 25, 2024, from healthline: <https://www.healthline.com/health/what-is-preventive-health-and-why-is-it-important>
- World Health Organization. (2024, 1 31). *Global Health Issues, Virtual Press Conference*. Retrieved 3 25, 2024, from World Health Organization: https://cdn.who.int/media/docs/default-source/documents/emergencies/transcripts/pdfvirtual-press-conference-on-global-health-issues31-january-2024.pdf?sfvrsn=6af2db9c_1

6. Appendix

Appendix 1: Variables of Diabetes Health Indicator Data

Variables	Description
Diabetes_012	0 = no diabetes 1 = diabetes
HighBP (Blood Pressure)	0 = no high BP 1 = high BP
HighChol (Cholesterol)	0 = no high cholesterol 1 = high cholesterol
CholCheck	0 = no cholesterol check in 5 years 1 = cholesterol check in 5 years
BMI	Body Mass Index
Smoker	Have you smoked at least 100 cigarettes in your entire life? [Note: 5 packs = 100 cigarettes] 0 = no 1 = yes
Stroke	(Ever told) you had a stroke. 0 = no 1 = yes
Heart Disease or Attack	coronary heart disease (CHD) or myocardial infarction (MI) 0 = no 1 = yes
PhysActivity	physical activity in past 30 days - not including job 0 = no 1 = yes
Fruits	Consume Fruit 1 or more times per day 0 = no 1 = yes
Veggies	Consume Vegetables 1 or more times per day 0 = no 1 = yes
HvyAlcoholConsump	Heavy drinkers (adult men having more than 14 drinks per week and adult women having more than 7 drinks per week) 0 = no 1 = yes
AnyHealthcare	Have any kind of health care coverage, including health insurance, prepaid plans such as HMO, etc. 0 = no 1 = yes
NoDocbcCost	Was there a time in the past 12 months when you needed to see a doctor but could not because of cost? 0 = no 1 = yes
GenHlth	Would you say that in general your health is: 1 = Poor 2 = Fair 3 = Good 4 = Very Good 5 = Excellent
MentHlth	How many days in the last month did you experience poor mental health, including stress, depression, or emotional issues?
PhysHlth	How many days in the last month did you experience poor physical health due to illness or injury?
DiffWalk	Do you have serious difficulty walking or climbing stairs? 0 = no 1 = yes
Sex	0 = female 1 = male
Age	<u>Age Category</u> 1) Age 18 to 24; 2) Age 25 to 29; 3) Age 30 to 34; 4) Age 35 to 39; 5) Age 40 to 44; 6) Age 45 to 49; 7) Age 50 to 54; 8) Age 55 to 59; 9) Age 60 to 64; 10) Age 65 to 69; 11) Age 70 to 74; 12) Age 75 to 79; 13) Age 80 or older
Education	<u>Education level</u> 1) Never attended school or only kindergarten 2) Grades 1 through 8 (Elementary) 3) Grades 9 through 11 (Some high school) 4) Grade 12 or GED (High school graduate) 5) College 1 year to 3 years (Some college or technical school) 6) College 4 years or more (College graduate)
Income	<u>Income scale</u> 1) Less than \$10,000 2) Less than \$15,000 (\$10,000 to less than \$15,000) 3) Less than \$20,000 (\$15,000 to less than \$20,000) 4) Less than \$25,000 (\$20,000 to less than \$25,000) 5) Less than \$35,000 (\$25,000 to less than \$35,000) 6) Less than \$50,000 (\$35,000 to less than \$50,000) 7) Less than \$75,000 (\$50,000 to less than \$75,000) 8) \$75,000 or more

Appendix 2: Variables for Medical Cost Personal data

Variables	Description
Age	Age of primary beneficiary
Sex	Insurance contractor gender: - Male - Female
BMI	Body mass index, providing an understanding of body, weight that are relatively high or low relative to height
Children	Number of children covered by health insurance / Number of dependents
Smoker	Smoking
Region	The beneficiary's residential area in US: - Northeast - Southeast - Southwest - Northwest
Charges	Individual medical costs billed by health insurance